China Ocean Eng., 2022, Vol. 36, No. 5, P. 827–838
DOI: https://doi.org/10.1007/s13344-022-0059-y, ISSN 0890-5487
http://www.chinaoceanengin.cn/ E-mail: coe@nhri.cn

Tidal Bore Dynamics Around the Similar Right-Angle Shoreline in the Qiantang Estuary, China

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Received May 19, 2022; revised September 2, 2022; accepted October 8, 2022

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Abstract

Tidal bores are a unique hydrodynamic phenomenon during flood tide in the Qiantang Estuary. The tidal bore propagation around the similar right-angle shoreline is rarely documented in tidal estuaries. To investigate tidal bores around this shoreline, a hydrodynamic model combined with a theoretical method is employed to reveal the characteristics of the bore propagation. The theoretical solution of the tidal bore intensity is deduced to illustrate the relationship of the incident tidal bores and the back-flow bores during the propagation. The hydrodynamic model based on shallow water equation is employed to perform the simulation of tidal bores in the estuary. Model results with respect to the bore height and the propagation speed of tidal bores have a favourable agreement with field data. The tidal bore dynamics in the neighborhood of the similar right-angle shoreline are elucidated. The characteristics of tidal bores in terms of water surface, velocity, bore steepness and the intensity are illustrated and the back-flow bore is analyzed by numerical and theoretical methods around the similar right-angle shoreline. The height of the back-flow bore relative to the incident tidal bore ranges from 1.05 to 1.77. Model result reveals that the ambient water depth and the shape of the similar right-angle shoreline are contributed to the back-flow bore formation.

Key words: back-flow bore, numerical model, shallow water equation, Qiantang Estuary, tidal bores

Citation: Wang, Q. S., Pan, C. H., Chen, F. Y., 2022. Tidal bore dynamics around the similar right-angle shoreline in the Qiantang Estuary, China. China Ocean Eng., 36(5): 827–838, doi: https://doi.org/10.1007/s13344-022-0059-y

1 Introduction

Tidal bore in the Qiantang Estuary is famous with its height and shapes around the world. Owing to the funnel-shaped estuary, the tide rushes into the Hangzhou Bay from the East China Sea and tidal bores are initially generated at Ganpu. Tidal bore progresses upstream more than one hundred kilometers at the speed of 20~30 km/h. The biggest tidal bores occur in the Yanguan Reach with a bore height of 3.0 m during spring tides (Pan et al., 2008). Tidal bores, especially the back-flow bores, attract millions of people watching. Meanwhile, tidal bore not only significantly impacts on sediment transport and riverbed morphology (Furgerot et al., 2016; Reungoat et al., 2017; Uncles et al., 2006; Wang and Pan, 2018; Xie et al., 2018), but also affects the safety of the old seawall. Many artificial structures have been built in the estuary to reduce these adverse effects by improving the flow condition (Cai et al., 2018; Zeng et al., 2017). The seawalls are distributed in the reach and prevent the attack from tidal bores. The forces and overturning moments on a vertical wall due to the reflection of undular bores and turbulent bores in a laboratory were studied (Ramsden and Raichlen, 1990; Ramsden, 1996). Mohapatra et al. (2000) analyzed the impact of tidal bore against a wall by employing a numerical model. Tidal bore dynamics in the unique shorelines induce the huge force implemented in the seawall. Therefore, it is indispensable to investigate the tidal bore dynamics around the shoreline in a tidal estuary.

Many researchers have dedicated to studying the process of the formation, development, propagation and dissipation of tidal bores (Bonneton et al., 2015, 2016; Filippini et al., 2019). Bonneton et al. (2015) investigated the formation of tidal bores dynamics in funnel-shaped estuaries by both a global tidal wave scaling analysis and field observations. To study the mechanism of tidal bore dynamics, many authors have been investigating the tidal bore dynamics in the laboratory (Chanson and Docherty, 2012; Chanson and Tan, 2010; Chanson and Toi, 2015; Docherty and Chanson, 2012; Huang et al., 2013; Leng and Chanson, 2017;

Foundation item: This work was financially supported by the National Natural Science Foundation of China (Grant Nos. 42276176, 51779228, and 41706099), the Zhejiang Province Public Welfare Technology Application Research Project (Grant No. LGF20E090003), and the Science and Technology Project of Zhejiang Water Conservancy (Grant No. RB2001 and RB2121).
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Rousseaux et al., 2016). Limited by the small-scale effect, reproducing tidal bores in a laboratory is defective in spite of the employment of novelty method of generating tidal bores (Rousseaux et al., 2016). Wolanski et al. (2004) studied the undular bore in the macro-tidal Daly Estuary and indicated that the tidal bore flows in natural estuaries differ significantly from those in the laboratory. Simpson et al. (2004) investigated the turbulence of tidal bore in the Dee Estuary. Xie and Pan (2013) studied the characteristics of tidal bores in the Qiantang Estuary based on field data. Additionally, the tidal bore is limited to be monitored and studied in a field site (Chanson and Docherty, 2012; Kawanisi et al., 2017; Tu et al., 2021; Zhu et al., 2012). Various refined models were developed to reproduce the small-scale characteristics of tidal bores (Berchet et al., 2018; Lubin et al., 2010; Ni et al., 2018), but it brings to the huge time-consuming computation for performing the simulation of tidal bore propagation in an estuary. Therefore, it is fundamental to establish an efficient model to investigate the tidal bores propagation.

Nonlinear shallow water equation is a sufficiently accurate and more efficient way to investigate turbulent bores. It is generally used to investigate the main physical processes in the estuarine system (Filippini et al., 2019). A surge, a rarefaction wave and tidal bore onset were simulated by nonlinear shallow water equation. The variation of the free water surface in an estuary can be captured via nonlinear shallow water equation. Based on the treatment of the pressure in the equation, the models are classified as hydrostatic or non-hydrostatic. The non-hydrostatic model denotes the inclusion of pressure in the governing equation, but the information about the non-hydrostatic component of the pressure field leads to a Poisson equation for the vertical velocity component (Antuono and Brocchini, 2013). The Poisson equation using the vertical velocity component is prone to instability because of the conflict between the Dirichlet condition at the free surface and the continuity equation. Subsequently, the semi-averaged models were developed to apply for the simulation of coastal dynamics (Antuono et al., 2017). Filippini et al. (2019) applied the fully nonlinear weakly dispersive equations to simulate the tidal bore propagation in a convergent estuary, verifying that the local dispersion effects do not influence the tidal bore onset. The nonlinear shallow water equations were suitable to simulate the propagation of tidal waves to bore formation in a small number of estuaries, such as the Qiantang Estuary (Münchow and Garvine, 1991). Breaking tidal bores in the Qiantang Estuary are prevailingly observed during equinox spring tides, and it is non-dispersive during the propagation. Numerical results by the shallow water equations are accurate compared with the results by the Boussinesq-type model (Kim and Lynett, 2011). Nonlinear advection in the shallow water equations was the main cause of bore development. Some researchers have shown that turbulent bores in the Qiantang Estuary can be reproduced via the non-dispersive shallow water equations with shock capturing methods (Madsen et al., 2005; Pan et al., 2007; Pan and Huang, 2010). This approach is used to simulate tidal bore propagation in the estuary with relatively good results. Consequently, numerical model of the tidal bore dynamics based on nonlinear shallow water equation was suitable to perform the simulation of the bore propagation in the estuary.

This study aims to carry out a study on the tidal bore dynamics around the right-angle shoreline in the Qiantang Estuary. With the blocking of the shoreline, back-flow bore occurs, but there is rare literature documented the characteristics of the tidal bore dynamics. In view of lacking the insight of the bore dynamics, a numerical model combined with a theoretical method is employed to investigate tidal bores around the similar right-angle shoreline. The data of tidal bores at field site are employed to present the bore dynamics. The theoretical analysis of tidal bores around the similar right-angle shoreline is presented in Section 2. In Section 3, tidal bores are reproduced by the numerical model, which is validated between model result and field data. In Section 4, the characteristics of tidal bores, and the temporal-spatial distribution of water surface around the shoreline are demonstrated. Bore height, bore steepness and the intensity are also calculated to illustrate the characteristics of tidal bores around the similar right-angle shoreline in Section 4. Finally, the conclusions are drawn in Section 5.

2 Theoretical analysis

2.1 Study area

The Qiantang Estuary is a 282 km long convergent estuary, located on the coast of the East China Sea, with the width decreasing from 98.5 km at the mouth to smaller than 1 km at the landward end (Xie et al., 2018). Fig. 1 shows the sketch of the Qiantang Estuary and locations of the relevant stations in this study. The photo of tidal bore propagation around the right angle shoreline in May, 2022 is provided by Zihang Huang (Fig. 1). Tidal bore is initially generated during flood tide at Ganpu and finally disappears at Wenyan

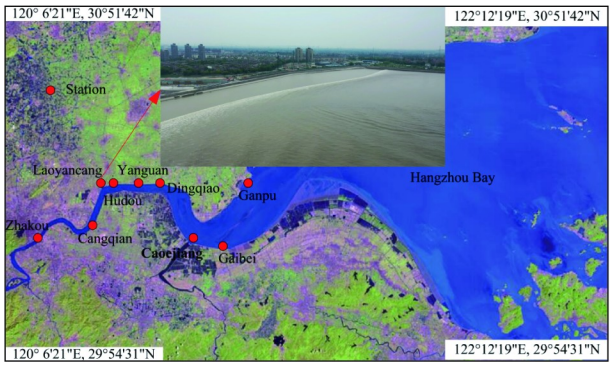

Fig. 1. Sketch of the Qiantang Estuary and locations of the relevant stations.

in the upper reach. The domain of tidal bore propagation is nearly 100 km over the estuary, which is controlled by both fluvial discharge and tide. Small river discharge was favourable to tidal range amplification and bore occurrence, while significant discharges offset the amplification mechanism related to estuary convergence. The direct effect of river discharge on tidal bore dynamics is neglected when the river discharge in the dry season is smaller than the mean annual discharge. The mean annual discharge of the Qiantang River is 1001 m^3/s based on the data from 1932 to 2021. The measured water levels from September 1 to 3, 2019 are shown in Fig. 2, including three tidal cycles and the specific discharge. The river discharge ranges from 525 to 1215 m^3/s during the hydrological survey, which has a negligible effect on tidal bores.

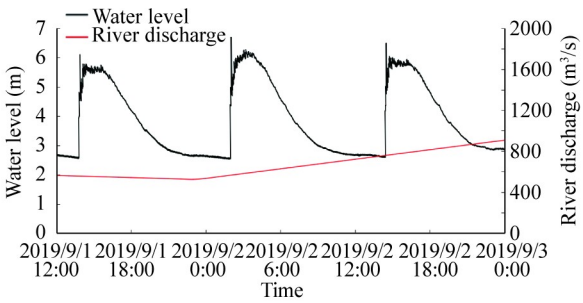

Fig. 2. Water levels at Laoyancang Station during the hydrological survey.

2.2 Tidal bores around the similar right-angle shoreline

Tidal range is locally amplified at Ganpu when the tide enters the estuary during flood tide. The tidal bore front becomes steeper until it forms tidal bore when the tide propagates over the estuary. After the formation of the bore, there is an abrupt rise in water depth at the tidal bore front (Chanson, 2011a).

The bore height is inconspicuous at the initial stage while significantly amplifies in the Yanguan Reach. The shape of tidal bores varies from Yanguan to Laoyancang. When the tidal bore progresses in the middle estuary, it yields the biggest tidal bores. Both line-shape and back-flow bores occur and the propagation patterns of tidal bore are diversified around the similar right-angle shoreline. The tidal bore front is gradually uniform, forming a famous line-shape bore owing to the relatively straight river channel in the reach. The tidal bores front spans the river, and then neatly advances to the upstream reach. Departing from the Yanguan Reach, tidal bores are suddenly blocked by the similar right-angle shoreline at Laoyancang. The shape of the tidal bore front is a function of its Froude number (*Fr*). The strength of tidal bore is proportional to the tidal bore Froude number. The sharp breaking front becomes very steep, resembling a water wall, and no secondary wave motions are present for $Fr>1.4$ (Leng and Chanson, 2015). The flow regime changes fundamentally and no longer has fluctuation characteristics. It is estimated that the tidal bore Froude number at Laoyancang Station during in-situ measurement in 2019 is 1.7, which is larger than 1.4. Consequently, breaking tidal bores are formed and rushed into the right-angle shoreline during astronomical spring tide. The incident tidal bores are reflected and thus the back-flow bore is generated around the right-angle shoreline (Fig. 3). The huge tidal bore attracts millions of people to watch this scenery every year. It is seen that tidal bore is strengthened around the shoreline due to the blocking of the shoreline.

2.3 Field measurement

The measurement of the tidal bore flows around the Laoyancang Station began from 11:00, September 1, 2019 during astronomical spring tide. Acoustic Doppler Current Profiler (ADCP) was used to sample the temporal and vertical variability of the tidal bore flows (Tu et al., 2021), and the field observation equipment of ADCP is displayed in Fig. 4. The special floating platform was installed with the ADCP (1200 kHz) to sample the tidal bore flows. The platform was fixed by throwing double anchors according to the station terrain and the propagation direction of tidal bore. The length of the anchor ranges from 5 to 8 times the water depth.

The data-sampling was conducted one hour prior to tidal bores, and was recorded every 2 seconds during the passage of tidal bores. The 2-second interval data sampling at Laoyancang was conducted during the arrival of tidal bores. It is deduced that the heights of the line-shape tidal bore and the

(a) (b)

Fig. 3. Photos of (a) the incident tidal bores and (b) the back-flow bores around the right-angle shoreline at Laoyancang.

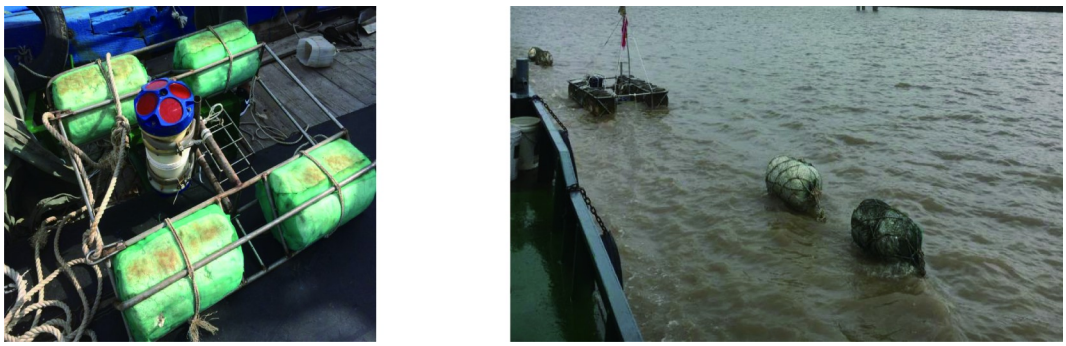

Fig. 4. Measurement of ADCP in field site.

back-flow bore approach 1.5 m and 1.7 m, defined by the variation of water level within 1 min (Fig. 5a).

The measurement of tidal bore flows with precision of 0.5° in direction and 0.25 cm/s in magnitude depends on the water depth, and the maximum velocity of 10 m/s can be recorded. The vertical six-point measurement is carried out when the water depth is larger than 4 m, which are namely measured in the surface layer (0.5 m below the surface), 0.2*D* (*D* is the water depth), 0.4*D*, 0.6*D*, 0.8*D* and the bottom layer (0.5 m above the bottom). The vertical three-point measurement is conducted when the water depth is smaller than 4 m, i.e. the surface layer, the middle layer (0.6*D*) and the bottom layer. When the water depth is smaller than 2 m, the one-point measurement is used, i.e. the middle layer (0.6*D*). The depth-averaged velocity is calculated by arithmetic average method among the vertical layers recorded by the ADCP in the field site with a distance of 800 m from Laoyancang. It is noted that there is a rapid change velocity during the passage of tidal bores, and the two peaks denote the velocities of the line-shape tidal bore and the back-flow bore (Fig. 5b).

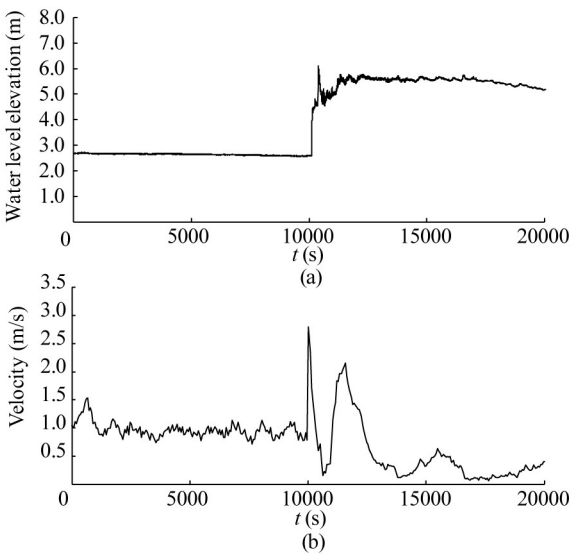

Fig. 5. Variation of (a) water level elevation and (b) depth-averaged velocity calculated by the vertical layers at the Laoyancang Station.

The tidal bores are generally intensified in terms of the bore height owing to the reflection of the seawall. It causes a large destructive force on the seawall nearby the shoreline. To present the dynamics of the bore propagation, numerical model combined with a theoretical method is employed to analyze the characteristics of tidal bore dynamics around the shoreline.

2.4 Theoretical equations

The flow properties in front of and behind the tidal bore front satisfy the continuity and momentum principles. The calculation of the propagation speed of tidal bore in a section was derived from the basis of the continuity and momentum principles (Chanson, 2011a; Lin, 2008; Pan et al., 2008; Xia, 1992). In this study, the propagation speed of the incident line-shape and back-flow bores around the shoreline by employing the continuity and momentum principles was deduced.

As the incident tidal bore travels around the shoreline (Fig. 6a), the bore front propagates towards the shoreline with a propagation speed U_1 (Fig. 6b). Blocked by the shoreline, the flow is generally stationary, i.e., v_1=0. Using the equations of conservation of mass and momentum gives the relationships between the flow properties in front of and behind the tidal bore front:

$$d_1(v_1 - U_1) = d_2(v_2 - U_1); \tag{1}$$

$$d_1 v_1(v_1 - U_1) + \frac{1}{2} g d_1^2 = d_2 v_2(v_2 - U_1) + \frac{1}{2} g d_2^2. \tag{2}$$

The combination of the above equations yields Eq. (3), and it is consistent with the previous studies (Berry, 2019; Chanson, 2011b; Lin, 2008; Pan et al., 2008; Xia, 1992).

$$U_1 = \sqrt{g\frac{d_2}{d_1}\left(\frac{d_2 + d_1}{2}\right)} = \sqrt{g d_2\left(1 + \frac{\eta}{2d_1}\right)}, \tag{3}$$

where v_1 and d_1 are the flow velocity and the water depth around the shoreline, v_2 is the flow velocity and it is positive downstream towards the shoreline, d_2 is the water depth of incident tidal bore, and g is the gravity acceleration. The incident tidal bore height η is defined by η=d_2−d_1.

Similarly, the incident tidal bore is reflected by the shoreline, and the back-flow bore front departs from the

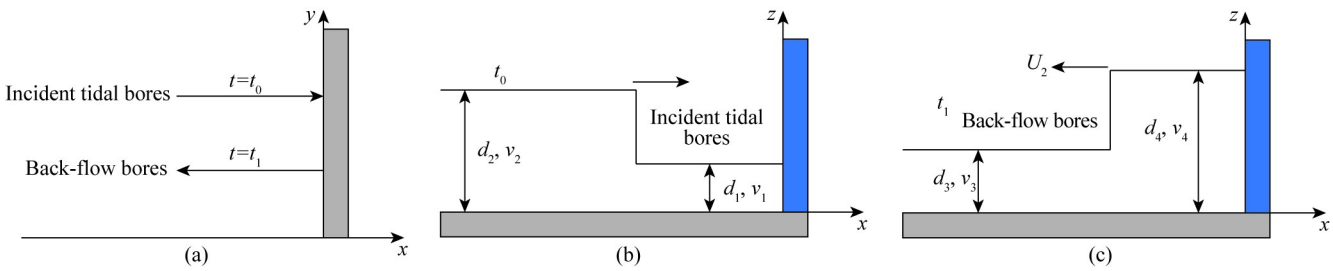

Fig. 6. Sketch of the tidal bore propagation around the shoreline.

shoreline zone with a propagation speed U_2 (Fig. 6c). In view of the stationary flow around the shoreline, the velocity is generally equal to zero, i.e., v_4=0. With the equations of conservation of mass and momentum, it yields the following result:

$$U_2 = -\sqrt{g\frac{d_3}{d_4}\left(\frac{d_3+d_4}{2}\right)} = -\sqrt{gd_3\left(1-\frac{\eta'}{2d_4}\right)}, \tag{4}$$

where v_3 and d_3 are the flow velocity and the water depth at offshore zone; v_4 and d_4 are the flow velocity and the water depth around the shoreline. The back-flow bore height η' is defined by $\eta'=d_4-d_3$.

The intensity of the incident tidal bore is defined by the following parameter:

$$\xi_0 = d_2/d_1 - 1. \tag{5}$$

The intensity of the back-flow bore is defined by the following parameter:

$$\xi_1 = d_4/d_3 - 1. \tag{6}$$

Assuming the steady-state flow, the water depth d_3= d_2 and the velocity v_3=v_2. Therefore, the following equation is deduced.

$$\xi_1^3(\xi_0+1)^2 + 2\xi_1^2(\xi_0+1)^2 - \xi_1\left(\xi_0^3+2\xi_0^2\right) - \xi_0^3 - 2\xi_0^2 = 0. \tag{7}$$

The relationship between the incident tidal bore intensity and the back-flow bore intensity is determined by the above equation.

3 Numerical model

On the basis of nonlinear shallow water equations, the numerical model of tidal bore propagation is developed. It is improved by the unstructured-mesh, Finite-Volume Coastal and Ocean Model. The hydrodynamic equations are shown in a detailed study (Chen et al., 2003; Huang et al., 2008). The improvement of numerical model of the Qiantang tidal bore was shown in previous studies (Cheng et al., 2017; Wang and Pan, 2017). Different turbulence models and the Smagorinsky parameter were tested to investigate the effect on the accuracy of numerical simulation. Different turbulence models have an effect on the accuracy of numerical simulation of the tidal bore flow compared with a negligible effect using different Smagorinsky parameters (Wang and Pan, 2017). It is revealed that the height of tidal bore and the velocity of the tidal bore flow in the $k-\varepsilon$ turbulence model is smaller than that in the Mellor-Yamada 2.5 turbulence model. The numerical result using Mellor-Yamada 2.5 turbulence model has a favourable agreement with field data. Consequently, the present model is closed mathematically using the Mellor-Yamada 2.5 turbulence model for vertical eddy mixing and the Smagorinsky parameterization for horizontal eddy viscosity. A sigma transformation in water depth is used to convert irregular bottom topography into a regular computational domain. The free water surface is integrated by solving vertically averaged equations with a smaller time step in the external model while the three-dimensional momentum equations are integrated with a larger time step in the internal model. Tidal bores were reproduced by the hydrostatic model and the non-hydrostatic model, and there was no difference of tidal bore onset between the hydrostatic model and the non-hydrostatic model. The numerical result of tidal bores by the hydrostatic shallow water equations was accurate with the result by the Boussinesq model based on a weakly non-hydrostatic pressure assumption (Kim and Lynett, 2011). Tidal bores were simulated by the hydrostatic shallow water equations with a satisfactory accuracy (Gusev and Lyapidevskii, 2005). In this study, the hydrostatic model is adopted to implement the investigation of the tidal bores propagation in the neighborhood of the similar right-angle shoreline in the estuary.

3.1 Model setup

The computational domain ranging from Ganpu to Zhakou is set up (Fig. 7). The horizontal domain is divided into 41221 elements and 21335 nodes by unstructured meshes. The horizontal resolution of the unstructured grid is 30 m around the similar right-angle shoreline in the Yanguan Reach. The water depth is configured on the basis of the bathymetric surveying with a scale of 1:10000. The domain is divided into 12 sigma layers in the water depth and thus the vertical resolution is approximately 1 m at Ganpu and increases to 0.1 m at Yanguan. The time steps of the external and internal models are 0.05 s and 0.5 s, respectively. The technique of parallel computing is employed in the simulation of tidal bores. Nine representative stations and a section in the numerical model are chosen to illustrate the characteristics of tidal bores around the similar right-angle shoreline (Fig. 7).

Nine representative stations were arranged in the zone of

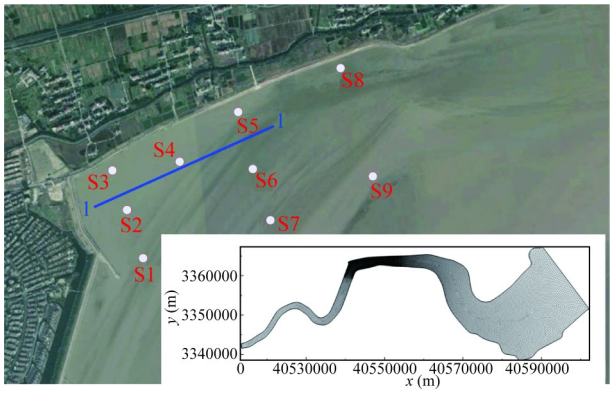

Fig. 7. Unstructured mesh in the computational domain and locations of representative stations and a section around the similar right-angle shoreline.

1800 m long and 600 m wide. Stations S1, S2 and S3 were located in the nearshore zone with a distance of 100 m from the upstream shoreline to illustrate the feature of back-flow bores nearby the shoreline, and Stations S5, S6, and S7 were located in the zone with a distance of 1200 m from the upstream shoreline to demonstrate the propagation of tidal bores along the longitudinal shoreline. Station S4 is an additional site between Stations S3 and S5 in the vicinity of the shoreline. Stations S8 and S9 were located in the zone with a distance of 1800 m from the upstream shoreline to display the range of back-flow bores propagation.

3.2 Model verification

In view of the occurrence of powerful breaking tidal bore, the field data in the Yanguan Reach during October 9, 2010 were employed to validate the numerical results. Numerical results are validated in terms of water level elevation and velocity. Herein, the bore height and propagation speed are compared with field data in the Yanguan Reach. Fig. 8 shows the verification of water level between field data and model result at Yanguan and Hudou stations. The 1-second interval model result is extracted to illustrate the abrupt rise of water level. It is seen that the simulated water level has a favourable agreement with the field data. Both model result and field data indicate a relatively obvious fluctuation of water level. The variation of suddenly-elevated water level is captured during the passage of tidal bores. The height of tidal bore is defined by the variation of water surface elevation within 1 min. It is deduced that the simulated bore height at Yanguan and Hudou stations is 3.02 m and 2.70 m, respectively.

To illustrate the variation of the velocity, a comparison of the velocity at the surface layer (0.5 m below water surface) and the bottom layer (0.5 m above river bed) between model result and field data is performed (Fig. 9). It is noted that the velocity is positive prior to tidal bores when the water column flows towards downstream. During the arrival of tidal bores, the velocity becomes negative and a reverse flow is generated. The rapid-change flow is captured during the propagation of tidal bores. The velocity has a significant increase during the arrival of tidal bores at Yanguan. The simulated peak velocity is basically equal to the measured data in spite of the fact that there are some fluctuated points between the model results and measured data. Departing from Yanguan, the velocity at the upper reach becomes smaller. The difference between the surface layer and the bottom layer is relatively noticeable prior to and after tidal bores, while the velocity becomes uniform at the moment of tidal bores arrival.

It is seen in Table 1 that the simulated propagation speed of tidal bore from Yanguan to Hudou is 5.79 m/s, which is slightly smaller than the measured data of 6.41 m/s. The relative error is 9.7%, and the simulated propagation speed of tidal bores matches with the field data. The calculated tidal bore Froude number in the reach is 2.05, indicating breaking tidal bores are formed during the propagation. The characteristics of tidal bores are reproduced in the Yanguan Reach. The back-flow bore is generated at Laoyancang, where tidal bores propagate towards upstream and meet at the similar right-angle shoreline. The tidal bore propagation around the similar right-angle shoreline is illustrated by the numerical model in the next section.

4 Results and discussions

4.1 Tidal bore height

The comparison of tidal bore height at the representative stations is displayed to illustrate the variation of tidal bores

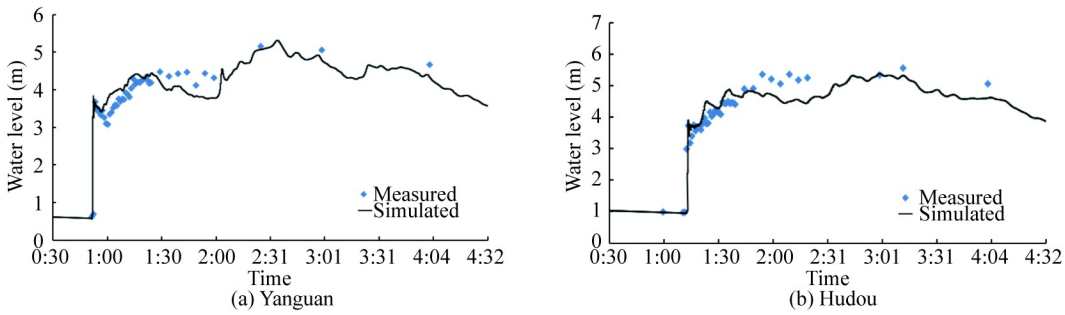

Fig. 8. Validation of water level elevation at (a) Yanguan and (b) Hudou stations during the passage of tidal bores.

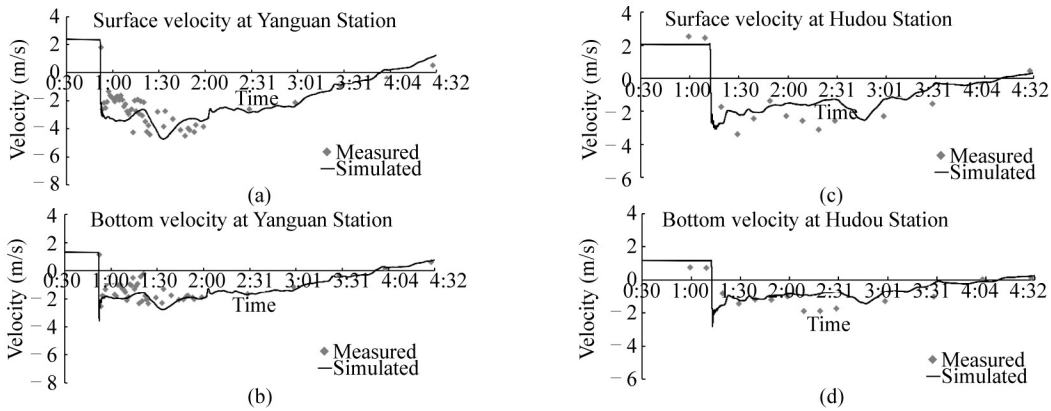

Fig. 9. Verification of the surface and bottom velocity at (a), (b) Yanguan and (c), (d) Hudou.

Table 1 Comparison of the characteristics of tidal bores

Characteristics		Yanguan	Hudou
Bore height (m)	Measured	2.99	2.75
	Simulated	3.02	2.70
Velocity (m/s)	Measured	4.52	3.40
	Simulated	4.76	3.08
Propagation speed (m/s)	Measured	6.41	
	Simulated	5.79	

during the propagation. Fig. 10 gives the simulated water level elevation with respect to the bore height at the stations in the neighborhood of the similar right-angle shoreline. It is seen that water level is suddenly elevated with the arrival of tidal bores. Moreover, there is a secondary elevation of water level after a few seconds. Owing to the blocking of the shoreline, tidal bores are reflected around the right-angle shoreline. Subsequently, the back-flow bore is generated and progresses towards downstream. The secondary height of tidal bore at the shoreline is significantly larger than the height of the incident tidal bores. The secondary height of tidal bore at the offshore stations is smaller than the bore height nearby the similar right-angle shoreline. The maximum bore occurs at the moment of the bores collision owing to the reflection of the shoreline. Water level increases to the highest elevation when the bore approaches the right-angle shoreline.

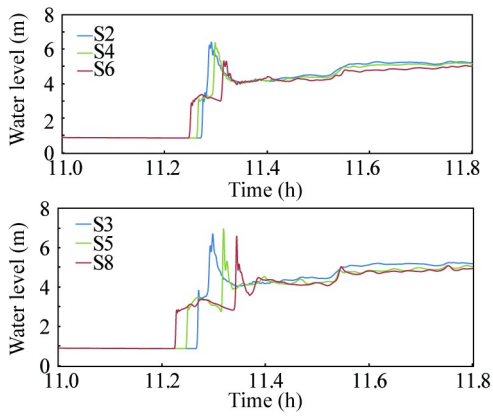

Fig. 10. Simulated water level elevation at representative stations around the shoreline during the passage of tidal bores.

In view of the reflection of the shoreline, the height of the back-flow bore has a significant increase. The simulated height of the back-flow bore relative to the incident tidal bore is calculated. It is noted that the relative height of the back-flow bore ranges from 1.05 to 1.77 at the stations around the shoreline (Fig. 11). In the offshore zone, the relatively mean height of the back-flow bore is 0.64, which is smaller than that of the incident tidal bore. It is indicated that the tidal bore height around the shoreline has a significant increase. The back-flow bore is generated around the shoreline and propagates towards downstream. The relative bore height near the shoreline is larger than that at the offshore zone. Back-flow bores have an overwhelming increase due to the reflection of the similar right-angle shoreline. The mean height of back-flow bore is equal to 1.40 times the incident bore at the corner during the tidal bore propagation. It is inferred that the shape of similar right-angle shoreline causes the accumulation of tidal prism with the arrival of tidal bores and the back-flow bore is significantly large relative to the incident tidal bores at the initial stage.

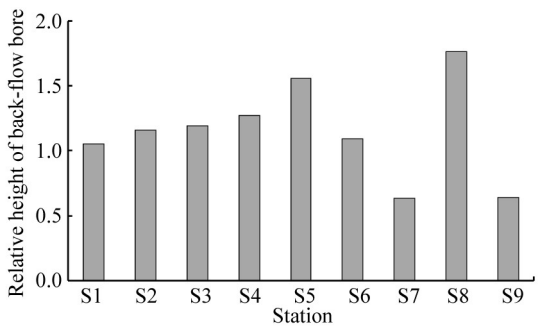

Fig. 11. Calculated relative height of back-flow bores.

4.2 Tidal bore intensity

To illustrate the variation of the tidal bore intensity, the comparison of the incident line-shape tidal bore and the

back-flow bore between theoretical solution and model result is shown in Fig. 12. Model results of the incident tidal bore and the back-flow bore intensity have a favourable agreement with the theoretical result although the tidal bore intensity at the offshore stations slightly deviates from the theoretical result. It is noted that the height of the back-flow bore is larger than that of the incident tidal bore around the shoreline, but the back-flow bore intensity is smaller than the incident tidal bore intensity. The back-flow bore intensity is close to the magnitude of the incident tidal bore intensity when the incident tidal bore intensity is smaller than 1. However, the back-flow bore intensity is equal to the half magnitude of the incident tidal bore intensity when the incident tidal bore intensity is larger than 1. Moreover, model results of the incident tidal bore and the back-flow bore intensity are smaller than the theoretical results. It is attributed to the fact that there is no boundary friction in the theoretical method. The back-flow bore intensity increases when the incident tidal bores become large, and this phenomenon is obvious in the neighborhood of the similar right-angle shoreline.

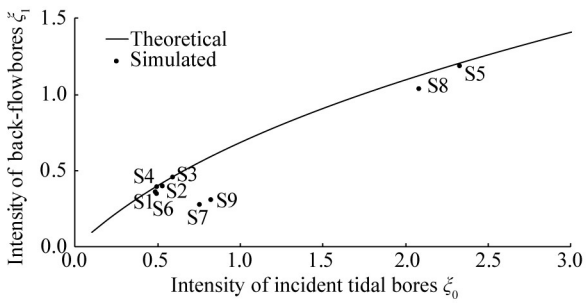

Fig. 12. Comparison of the incident tidal bore and the back-flow bore intensity between theoretical and simulated results.

The critical factors affecting tidal bores are the ambient water depth and the shape of the shoreline. It is observed that the bore height is larger at the shoreline than that at the offshore zone. The shallow water depth indicates a large tidal bore occurrence, and vice versa. Moreover, the flow is nearly stationary in the neighborhood of the similar right-angle shoreline, and tidal bore becomes large in these zones when it travels to the relatively stationary flow. Additionally, the tidal bore intensity is affected by the shape of the shoreline. It is noted that tidal bores become weak in the zone far away from the corner. However, the back-flow bore is obviously intensified at the corner of the similar right-angle shoreline. It reveals that the tidal bore intensity is related to water depth and the shape of the shoreline. The maximum simulated tidal bore intensity was smaller than the maximum theoretical intensity.

4.3 Tidal bore velocity

The variation of the simulated velocities at the surface layer during the passage of tidal bore is manifested in Fig. 13. The velocity is obviously fluctuated with the arrival of tidal bores. It is observed that the initial velocity is nearly equal to zero around the shoreline but has a significant rise induced by the back-flow bore. The combination action of the incident tidal bores and the back-flow bores makes the velocity fluctuation remarkable. The velocity profiles along the selected section (shown in Fig. 7) are depicted in Fig. 14. It is seen that the velocity profile has a spatial difference along water depth. There is a logarithmic distribution of the velocities along water depth. The water column at the upper layer has a large velocity before approaching the shoreline. At the moment of tidal bore collision, the tidal bore flow becomes weak and the water level has a remarkable uplift around the shoreline. There is a mutual conversion between tidal bore kinetic energy and potential energy during the propagation of tidal bores. A rapid-moving flow propagates towards upstream with the arrival of tidal bores relative to the stationary flow around the shoreline. As the tidal bore approaches the right-angle shoreline, the flows are reversed to propagate towards downstream with the reflection effect by the shape of the shoreline. The reversed flow counteracts the propagation of the incident tidal bore and becomes weak, blocked by the similar right-angle shoreline. The velocity significantly varies at different moments during the propagation of tidal bore. The back-flow bore has a reverse effect on the velocity magnitude and makes the velocity decreased around the shoreline. Additionally, it is noted that the flows in front of back-flow bores move upstream, where it is dominated by the incident line-shape tidal bores. The horizontal velocity in the vicinity of back-flow bores becomes weak and the vertical velocity is enhanced, flowing upstream in the upper layer and moving downstream in the lower layer of water column. It is indicated that the incident line-shape tidal bore is more powerful than the back-flow bores in this selected section. The phenomenon is consistent with the measured data nearby the section (Figs. 9c and 9d) and the simulated result at Station S5 (Fig. 13).

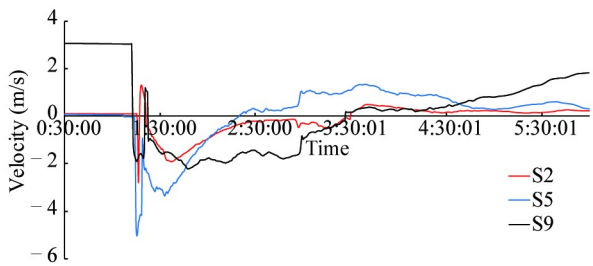

Fig. 13. Variation of the simulated velocities at the surface layer during the passage of tidal bores.

The comparison of the propagation speed between the incident line-shape bore and the back-flow bore is shown in Fig. 15. The calculated propagation speed of tidal bores is obtained by the distance between Yanguan and Laoyancang dividing the time interval of the tidal bore arrival. The simu-

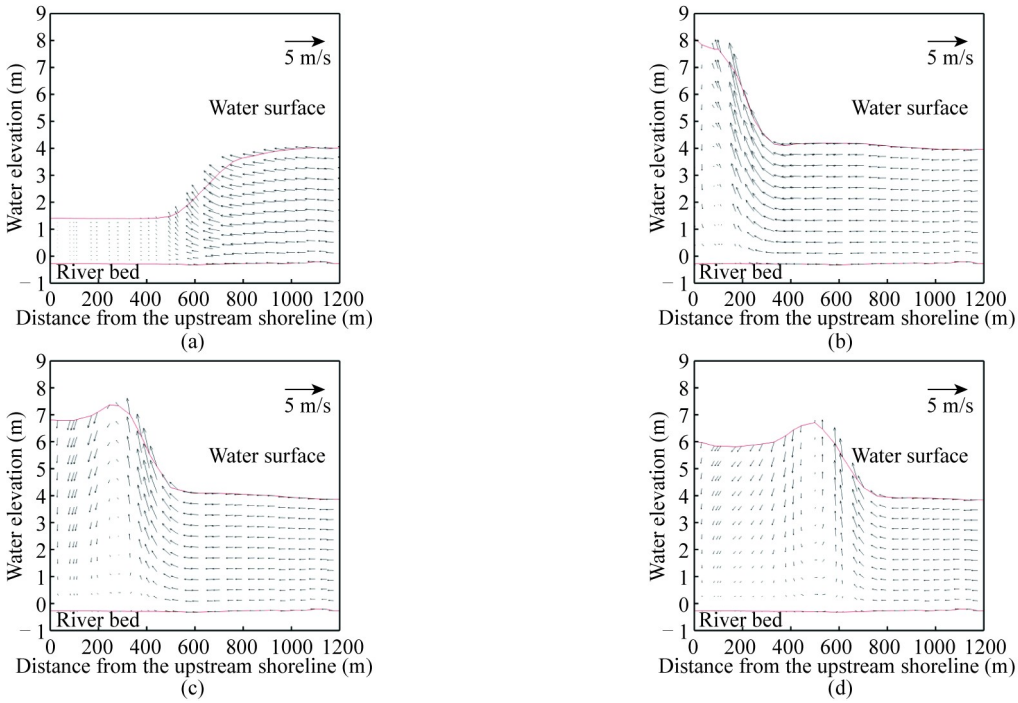

Fig. 14. Velocity profiles during the passage of tidal bores.

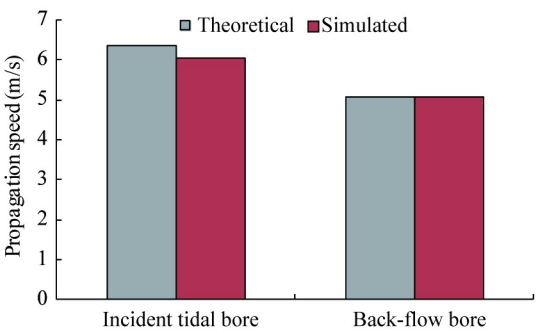

Fig. 15. Comparison of the propagation speed of tidal bores.

lated propagation speed of the line-shape bore and the back-flow bore has a satisfactory agreement with the theoretical result. The propagation speed of the back-flow bores is smaller than that of the incident tidal bores when the height of the back-flow bores is larger than that of the incident tidal bores. It is inferred that the propagation speed of the back-flow bores is equal to 80% of the propagation speed of the incident tidal bores. The back-flow bores play a reverse role in the propagation of tidal bores and the height of back-flow bores is larger than that of the incident tidal bores.

4.4 Bore steepness

Bore steepness is a critical parameter to evaluate the tidal bore intensity. It is defined by the ratio of the tidal bore height relative to the bore length along the longitudinal channel (Chanson, 2011b). The steeper bore front indicates a larger tidal bore intensity during the tidal bore propagation. Zhou et al. (1992) analyzed the field data of tidal bores during August 24 to October 24, 1968 and found that the bore steepness ranged from 0.1 to 0.34 in the Qiantang Estuary. Xie et al. (2012) deduced the bore steepness ranging from 0 to 0.35 by employing the measured data in Jiashao bridge during astronomical spring tide in October, 2011. Berry (2019) established the formula of bore steepness in a minimal model for tidal bore. Huang et al. (2013) derived a calculation formula of bore steepness by using the one-dimensional mass and momentum equations, and calculated the steepness in a laboratory. Zhang et al. (2020) used the steepness formula to calculate the bore steepness at Yanguan station. Viero and Defina (2018) proposed a criterion to predict the bore formation and established the relationship of the bore steepness as a function of the tidal bore Froude number. Filippini et al. (2019) analyzed tidal bore formation in convergent estuaries and the criterion to determine tidal bore onset by the bore steepness. Ningsih et al. (2019) calculated different bore steepnesses and found that the bore steepness decreases with the increasing initial water depth and the breaking bore has a large steepness.

In this study, the analytical algorithm derived from the equations of mass and momentum is used to calculate the bore steepness. It is calculated by:

$$\frac{\partial \zeta}{\partial x} = \left(\frac{d}{u}\frac{\partial u}{\partial t} - \frac{\partial \zeta}{\partial t}\right) \Big/ \left(-\frac{gd}{u} + u\right), \tag{8}$$

where ζ is the water surface elevation, d is the water depth, and u is the velocity along the longitudinal channel.

The calculated result of the bore steepness at Station S2

around the shoreline during the propagation of tidal bore is shown in Fig. 16. It is seen that there are abruptly fluctuated peaks and troughs during the passage of tidal bores. Prior to the incident tidal bores, it is smaller than 0.001 and implies a calm water surface in the zone. The passage of tidal bore was associated with an increase in the bore steepness (Bonneton et al., 2015). The bore steepness is larger than 0.001 when a tidal bore occurs, and it is consistent with the conclusions in previous studies (Filippini et al., 2019; Viero and Defina, 2018). Subsequently, it suddenly rises to the peak with the arrival of tidal bores. A violent water surface is generated and the turbulence of tidal bore is intensified. After a few seconds, a secondary bore is formed due to the reflection of the shoreline, and the bore steepness drops to the trough. The incident tidal bores and the back-flow bores induce the relatively sharp fluctuation, and lead to the occurrence of the large bore steepness. Departing from tidal bores, the violent water surface becomes relatively calm and the bore steepness is slightly small. The bore steepness is closely related to the water level fluctuation during the passage of tidal bores. The water level rapidly rises several meters in a short time and the bore steepness increases sharply at the moment of the arrival of tide bore. There are still large fluctuations in the water surface within more than ten minutes after tidal bores. The calculated bore steepness ranging from 0 to 0.24 is close to the steepness in the field site (Zhou et al., 1992).

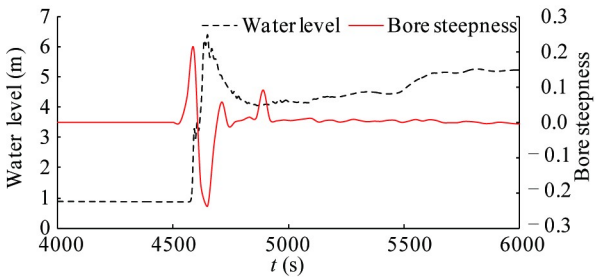

Fig. 16. Calculated bore steepness during the passage of tidal bores.

4.5 Water surface profiles

The temporal changes of water surface in the neighborhood of the similar right-angle shoreline are illustrated in Fig. 17. It is observed that water surface is initially low around the shoreline before the arrival of line-shape bores (Fig. 17a). Water surface profiles in the river channel are not uniform due to the ambient bathymetry. At the moment of the bores collision, there are large congregations of water column at the corner (Fig. 17b). Finally, the line-shape tidal bores and the back-flow bores propagate towards upstream (Fig. 17c and Fig. 17d). Water surface undergoes a drastic rise with the arrival of tidal bores. It is seen that the thick water surface profiles indicate an intensive variation of water surface. Water surface increases to a large amplitude and reflects to form a secondary bore around the similar right-angle shoreline. When the back-flow bore is generated and propagates towards downstream, water surface fluctuates obviously and gives a secondary rise.

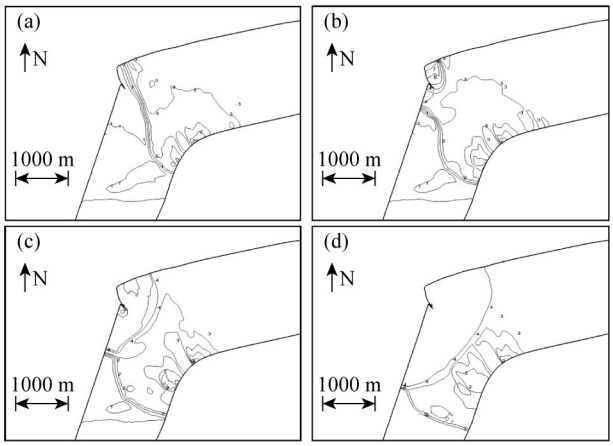

Fig. 17. Distribution of water level profiles during the passage of tidal bores (unit: m).

5 Conclusions

The tidal bore propagation around the similar right-angle shoreline is demonstrated by the theoretical analysis and the numerical model. The tidal bore intensity and the propagation speed of tidal bores are theoretically deduced and the back-flow bores are numerically reproduced. The following conclusions are drawn.

(1) The relationship between the incident tidal bores and the back-flow bores during the propagation is demonstrated by the theoretical solution of the tidal bore intensity. The back-flow bore intensity is smaller than the incident tidal bore intensity although the height of back-flow bore is larger than that of the incident tidal bore around the similar right-angle shoreline. Meanwhile, the propagation speed of the back-flow bores increases with the amplification of the intensity. The back-flow bore intensity is close to the magnitude of the incident tidal bore intensity when the incident tidal bore intensity is smaller than 1. However, the back-flow bore intensity is equal to the half magnitude of the incident tidal bore intensity when the incident tidal bore intensity is larger than 1.

(2) The model based on nonlinear shallow water equation is an efficient way to test the tidal bore propagation in the estuary. Model results of tidal bores in the Yanguan Reach have a favourable agreement with field data. The similar right-angle shoreline makes tidal bores congested around the corner and thus induces the back-flow bore occurrence. The height of back-flow bore relative to the incident tidal bore ranges from 1.05 to 1.77. The relative height of the back-flow bore is related to the shape of shoreline and the ambient water depth.

(3) Water surface undergoes a drastic elevation with the arrival of tidal bores around the shoreline. Tidal bores are

blocked to form a back-flow bore around the similar right-angle shoreline, propagating towards downstream. The propagation speed of the back-flow bore is equal to 80% of the propagation speed of the incident tidal bores. The back-flow bores play a reverse role in the propagation of tidal bores. Moreover, bore steepness fluctuates abruptly during the passage of tidal bores. The maximum calculated bore steepness is 0.24, which is close to the steepness in field site, indicating an intensive breaking tidal bore occurrence around the shoreline.

(4) The study of tidal bore dynamics around the shoreline provides a basis for investigating the interactions associated with tidal bores and the seawall. Nevertheless, the tidal bore dynamics around the shoreline are associated with structures, sediment transport and riverbed change. A Boussinesq-type model will be further conducted to study the small-scale mechanism of tidal bore dynamics around the shoreline.

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